K Madhumitha

Full-Stack Developer | AI/ML Developer | Computer Vision with Deep Learning

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PROFILE

Computer Science student with a strong interest in full-stack development, artificial intelligence, and computer vision. Experienced in building FullStack applications and implementing machine learning models using Python, React, SQL, and modern web technologies.

EXPERIENCE

Indian Space Research Organization (ISRO) – *Research Intern* *Apr. 2026 – May 2026*

- Researched **vision-based satellite pose estimation** techniques and studied deep learning approaches for satellite detection, localization, and computer vision applications.
- Reviewed and analyzed **deep learning research papers** on object detection, satellite imagery, and vision-based pose estimation to understand current approaches and model architectures.
- Implemented and trained **YOLOv8 and RF-DETR** models on satellite image datasets for object detection and comparative performance analysis.
- Validated model performance using relevant **object detection evaluation metrics** and compared YOLOv8 and RF-DETR based on their detection performance across the dataset.
- Conducted **experimental analysis of detection models** to identify differences in model performance and understand the applicability of modern deep learning architectures to satellite vision tasks.

PROJECTS

Driver Drowsiness Detection *Mar. 2026 – May. 2026*

Python, OpenCV, Computer Vision

- Developed an **AI-powered drowsiness detection system** using facial landmarks and Mouth Aspect Ratio (MAR) to detect yawning in real time.
- Implemented **real-time audio alerts** and visual monitoring to improve driver safety under varying lighting conditions.

NovelNest *Dec. 2025 – Jan. 2026*

React.js, Tailwind CSS, Supabase, Row Level Security (RLS)

- Built a **full-stack multi-user content platform** with secure authentication, role-based access control, and approval workflows.
- Implemented **Supabase cloud database and Row Level Security (RLS)** for secure, real-time data management and controlled user access.

Expense Tracker *Sep. 2025 – Nov. 2025*

MongoDB, Express.js, React.js, Node.js, JWT, Chart.js, Tailwind CSS

- Developed a **full-stack personal finance management platform** with JWT authentication, multi-account transaction tracking, and recurring payment management.
- Built **interactive financial analytics and budgeting features** using Chart.js, including spending insights, budget alerts, and transaction management.

EDUCATION

Bachelor of Engineering (B.E.) in Computer Science and Engineering
Cambridge Institute of Technology, Bangalore *2023 – 2027*
CGPA: 8.2 / 10.0

TECHNICAL SKILLS

Languages: JavaScript, Python, Java, SQL, HTML, CSS, Kotlin
Frameworks & Libraries: React.js, Node.js, Express.js, YOLOv8, RF-DETR, OpenCV, FastAPI
Databases: Supabase, MySQL, MongoDB, IndexedDB
Developer Tools: Git, GitHub, Netlify, Canva, Jupyter Notebook
AI/ML: Deep Learning, Computer Vision, TensorFlow, Keras, PyTorch, Scikit-learn

CERTIFICATIONS

Data Science using Python – Infosys Springboard
Frontend Development – Infosys Springboard
Programming in Java – (Elite) NPTEL
Introduction to Computer Vision With Deep Learning – CIT (CGEP)